Why fear an artificial mathematician?

Would a math AI want to break out?

What kind of math AI are we interested in talking about?

- Let’s imagine that we somehow create an AI which is really good at math.
- A bit more concretely, let's imagine it proves/disproves most currently central conjectures (the Riemann hypothesis, P vs NP, Navier–Stokes, etc.).
- This probably means that it makes at least as much progress as the human mathematical community would make in 200 years.
- In fact, it can probably successfully pursue mathematical goals over very long intervals of its “subjective time”.

What might such a math AI be up to more precisely? pt 1/3

- we should probably imagine it doing a bunch of most of the following:
  - defining/constructing objects/types/structures (inventing real numbers, groups, manifolds, etc.)
  - defining properties (primality, continuity) and relations (divisibility, being a subspace)
  - defining functions (the sum of two numbers, the limit of a sequence, the derivative of a function)
  - developing new mathematical frameworks (e.g. figuring out how to do algebraic topology category-theoretically and using homological algebra)
  - coming up with and developing mathematical ideas (e.g. the idea of a probabilistic construction)
  - working out examples, computing stuff out
  - solving problems
  - proving theorems
  - coming up with good problems
  - coming up with good theorems and conjectures
  - coming up with games and playing games
  - coming up with algorithms
  - writing code (possibly including doing ML)
  - also, various loose/heuristic/physicsy versions of these things
- in other words: it'll be thinking about a bunch of mathy stuff

What might such a math AI be up to more precisely? pt 2/3

- it will probably also be doing thinking somewhat like the thinking a human does when asking some of the following questions:
  - what auxiliary construction would help with this problem?
  - which techniques could work here?
  - what would be useful to figure out to solve this problem?
  - what was the main idea of this proof?
  - is this a good way to model the situation?
  - can I explain that clearly?
  - what caused me to be confused about that?
  - why did I spend so long pursuing this bad idea?
  - how could I have figured that out faster?
  - which question are we asking, more precisely?
  - why am I even interested in this problem?
  - what is this analogous to?
  - what should I read to understand this better?
- in other words: it'll be thoughtfully improving its mathematical thinking
  - arguably: it will be thinking about how to think better mathematically

What might such a math AI be up to more precisely? pt 3/3

the AI mathematician might also be thinking about:

- how it works / its own psychology
  - for example, it will be tracking what kinds of mathematical skills it has
  - it can do psychology research on itself
  - it can probably figure out what the rules of its boxed world are (possibly not fully, but to some meaningful extent)
  - actually, the stuff on the previous slide was already it thinking about how it should work
- how to allocate resources
  - for example, it could create some internal prediction markets and traders to be better able to track which proof attempts are promising
- various philosophical questions
  - why? because philosophical thinking is useful and interesting
    - eg all/most human sciences have come from philosophy
    - maybe of particular interest in this context: logic, probability theory, computability and complexity theory, combinatorial game theory, game theory, information theory
- a bunch of other stuff, some of which would be quite alien to us

The questions we want to ask, more precisely

we are interested in a cluster of questions:

- Would the AI want to take actions that have big real-world effects?
- Would it be actively trying to come up with ways to break out?
  - Or is it more like: it would break out if it could, but it isn't putting in any effort?
- Are there any real-world things that the AI would want to do?
  - instrumentally? terminally?
- Could we present an AI with a button that does some real-world thing such that it would want to press that button?
- If the AI could send a copy of itself to a different data center, would it do that?
- Would the AI be interested in getting more compute for itself by hacking into another data center?
- Would it want to build more data centers for itself? Would it want to harvest more energy? Would it want to "eat" the Sun?
- Would it want to control whatever aspects of its own world are not already controlled by it?
  - example: Would it want to change what it is being trained to do?
  - example: Would it want to be more in control of what problems it is solving?
- Would it want to ensure it is not shut down?

Wait, why should we care about this question?

- it would be nice to understand this question for understanding how (not) to make a good/safe AGI
  - if even a math AI would totally want to break out, that's a reason for pessimism
  - it's roughly equivalent to the same question for other AIs trained purely myopically to output solutions to some other "abstract problems", eg prediction problems
  - it's relevant to making a task AI in general

(also, it's just a good philosophical/scientific question imo)

a counter: “But the AI will only care about some math stuff!”

- “Okay, maybe we did eg RL to make the math AI, so it will care about something. I’m even willing to admit it will have sth like:
  - mathematical interests;
  - a sense of what to think about (at least when asked to prove something);
  - an ability to set itself mathematical goals, like to try to prove some lemma on the way to proving a theorem;
  - an ability to set itself vaguer mathematical goals, like to understand/develop some technique;
  - other sorts of mathematical values (mathematical virtues? mathematical principles?).”
- “But it will only care about some math stuff!”
- “Why would it develop values that relate to the real world? How could it?”
- “At no point in training does caring about the real world get reinforced!”

Some analogies

You might try this argument on:

- an AI (with actuators) trained to do science on an island
    - will it be interested in doing off-island stuff?
- an AI (with actuators) trained to do science on a planet
    - will it be interested in doing off-planet stuff?
- an AI (with actuators) trained in a fairly carefully sealed physical lab
    - will it be interested in doing stuff outside the lab?
- imagine that our universe is simulated on some computer embedded in a larger universe
    - would we want to break out?

What’s going on in these situations?

- instrumental reasons
  - you can do stuff in the real world that affects stuff you already cared about
- values just keep giving action-guidance in new contexts
  - it’s not like if you grew up on Earth, you now have no sense of what to do on Mars
- in particular, values about what to be up to generalize, and values about what to be like generalize

argument: these also apply to the math AI case

a counter: “Wouldn’t this prove way too much?”

- “Surely AlphaZero doesn’t want to break out in any meaningful sense?”
- “We could maybe imagine a chess AI that wants to break out, but it’s apparently easy to make one that doesn’t? So why wouldn’t it be easy to make a math AI that doesn’t either?”
- “You minimally owe me some account of the difference between a math AI and a chess AI.”

important differences:

- AlphaZero is missing intellectual capabilities
    - There is probably no sense in which it could understand what any real-world button does.
    - AlphaZero can’t set itself targets in any large space.
    - AlphaZero doesn’t have any mental loop that’s like “oh, X is good for Y? ok, then I will do X!” for any meaningfully large space of Xs
    - AlphaZero isn’t creative, and it can’t learn to play arbitrary games.
- roughly speaking, AlphaZero’s search is largely restricted to chess trajectories, whereas the math AI will effectively be able to run arbitrary searches

Further questions

- When would the AI break out? Like, before proving that P ≠ NP, or after?
  - Obviously, this depends on how tightly the AI is boxed.
  - but like: Does the AI break once breaking out becomes "objectively easier" than proving the theorems it is proving, or does it have a large skill gap here?
- Suppose the AI breaks out, gets more compute or whatever, and proves the theorem it was locally asked to prove. Does it stop then?
  - my view: strong no
- More generally, what will the AI do after breaking out?
  - my view: all sorts of wild stuff; anyway, this doesn't go well for humans
- What if the AI is trained against breaking out?
  - my view: this seems really difficult
- What if the AI is boxed very tightly except for its output — it only gets to interact with the outside world by outputting something — and what if further it is somehow trained to output short proofs/constructions, and then at some point deployed so now its outputs are used for doing some real-world thing?
  - I think this also ends poorly for humans, but this deserves another long discussion.

Some general intuitions

- There's a "pile of gold" out there in the real world for the math AI.
    - whether it gets there is a somewhat different question
- Not caring at all is a sharp/strange/"measure 0" property.
- The math AI (that actually works) will be creative, and it will think outside the box.
    - There is imo the following common mistake here: imagining a safe math AI, but that AI also not getting anywhere actually.
- A sort of relentlessness is to be expected. The math AI is already in the business of coming up with very difficult things to get around obstacles
    - figuring out a way to break out is a lot like coming up with a very useful concept or proving a very useful theorem

Thanks for listening!

and thanks for discussions and ideas:

- Tsvi Benson-Tilsen
- Lucius Bushnaq
- Jake Mendel
- Samuel Buteau
- Matt MacDermott
- Oliver Richardson
- Simon Skade
- Sam Eisenstat